

Solution Requirements

Date	02 July 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Current version does not require user login. Future versions may support secure user authentication for personalized experiences.	Low
2	Authorization levels	Current version provides a single user role with access to all available feature. Future versions may include Admin and User roles.	Low
3	External interfaces	The system integrates with Google Gemini API for recommending facts and suggestions and Hugging Face Transformers for NLP-	High

		based event analysis.	
4	Transactions processing	The system processes user inputs (their study category and prompt), analysis their emotions and provides suggestions, it also stores the conversation history.	High
5	Reporting	The system analysis the emotion of the user and provides a polite suggestion to overcome the issue the user is facing in their learning journey.	High
6	Business rules, etc.	User provide both the category of their study and their input prompt. System analyzes the input prompt and understanding the dominating emotion along with a suggest to help the user.	Medium
7	Compliance to laws or regulations	The application does not collect sensitive personal information. It follows general data privacy principles while storing user interactions.	High
8	<i>Other</i>	System stores conversation history and user emotion history for future reference.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system analyses the emotions and generates a suggestion and tip quickly.	Average response time < 2 seconds
2	Scalability	The system should support additional users and future cloud deployment.	Supports at least 100 concurrent users in future deployment.
3	Security & Data Privacy	User data, user interactions should be securely handled.	No sensitive user data stored; secure API communication.
4	Reliability & Availability	The application should operate continuously without major failures.	System availability > 95% during testing.
5	Usability & Accessibility	The interface should be intuitive, user-friendly, and easy to navigate.	Users can generate suggestions and get analysis of the emotions with just their user prompts.
6	Maintainability & Portability	The code should be modular, reusable, and easy to deploy across different environments.	Modular architecture with reusable services and documented APIs.